MODULE 1

FUNDAMENTALS OF CLOUD COMPUTING

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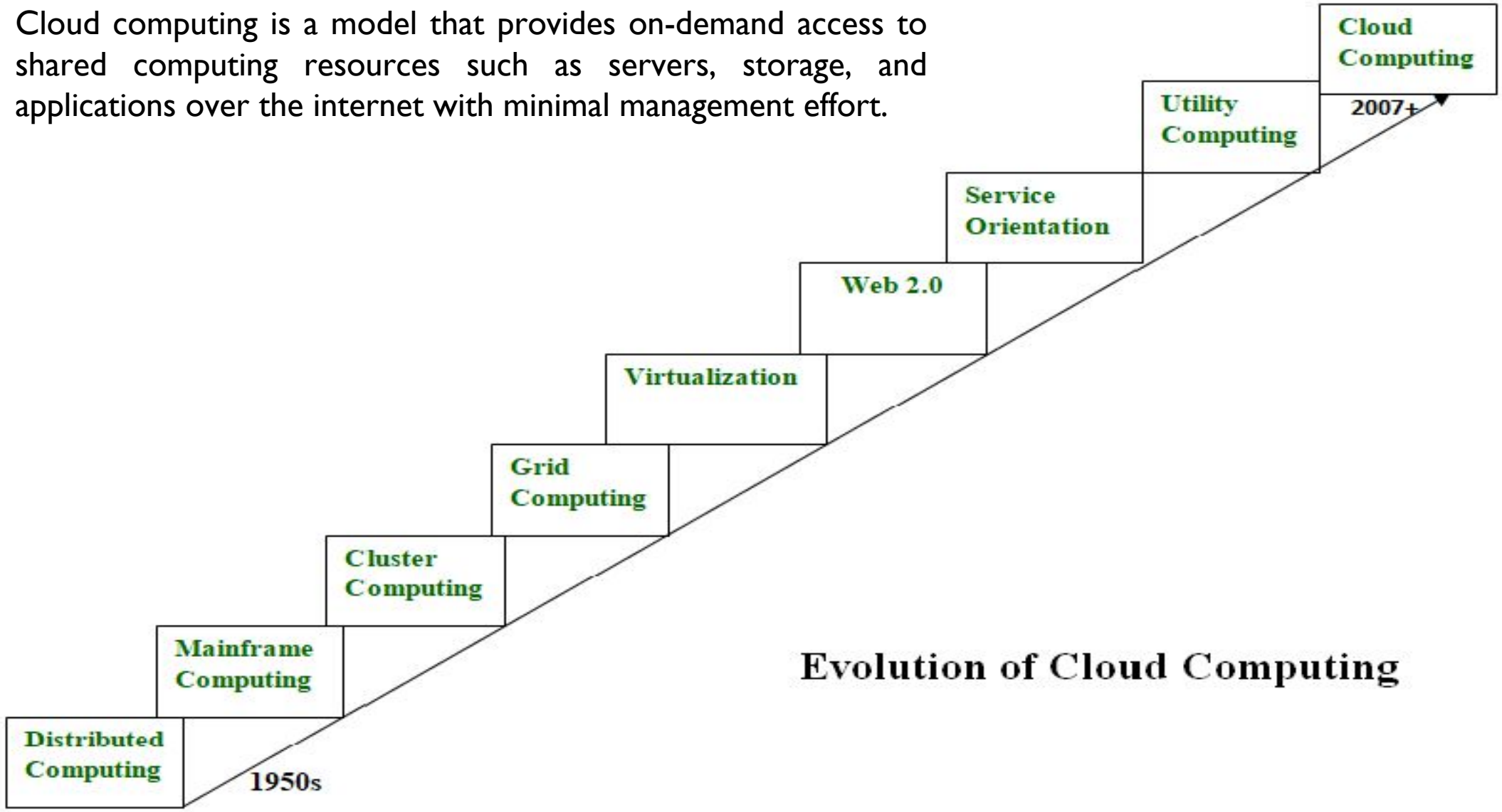
HISTORY OF CLOUD COMPUTING

Cloud computing is a model that provides on-demand access to shared computing resources such as servers, storage, and applications over the internet with minimal management effort.



Distributed Systems

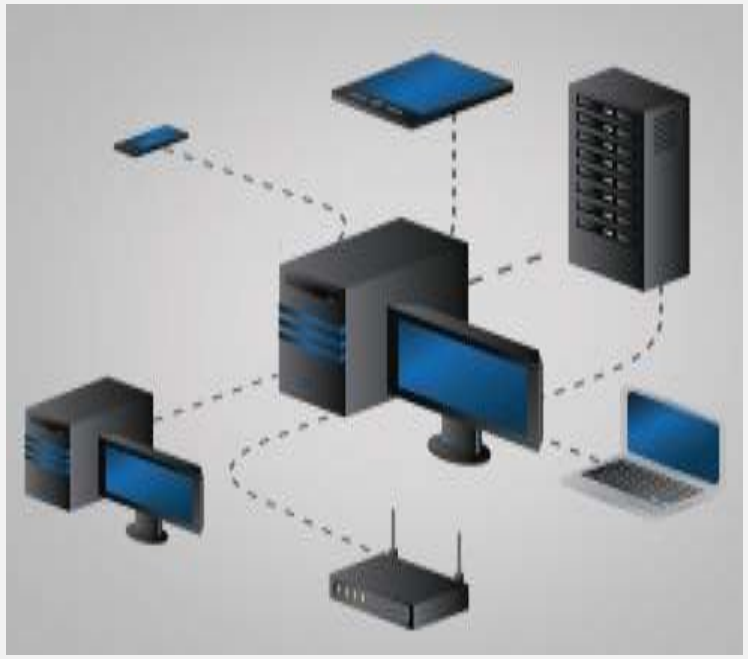
- Itis composition of multipleindependent systems but all of them are depicted as a single entity to the users.
- Purpose of distributed systems is to share resources and also use them effectively and efficiently.
- Characteristics:
- Scalability
- Concurrency
- continuous availability
- heterogeneity, and
- independence in failures.
- But main problem was that all systems were required to be present at same geographical location.

Mainframe computing

- Mainframes first came into existence in 1951 are highly powerful and reliable computing machines.
- Capable of handling large data such as massive I/O operations.
- Even today these are used for bulk processing tasks such as online transactions etc.
- These systems have almost no downtime with high fault tolerance.
- But these were very expensive.
- To reduce the cost, cluster computing came as alternative to mainframe technology.

Cluster computing

- In 1980s, cluster computing came as alternative to mainframe computing.
- Each machine in cluster was connected to each other by network
  with high bandwidth.
- These were way cheaper than mainframe systems. These were equally capable of high computations.
- Also, new nodes can be added easily to cluster if it was required.
- Thus, problem of cost was solved to some extent but problem related to geographical restrictions still pertained.
- To solve this, concept of grid computing was introduced.



Grid computing

- In 1990s, concept of grid computing was introduced.
- Different systems were placed at entirely different geographical locations and these all were connected via the internet.
- These systems belonged to different organizations and thus grid consisted of heterogeneous nodes.
- Although it solved some problems but new problems emerged as the distance between nodes increased.
- Main problem which was encountered was low availability of high bandwidth connectivity and with it other network associated issues.
- Thus, cloud computing is often referred to as “Successor of grid computing”.

Virtualization

- It was introduced nearly 40 years back.
- It refers to process of creating virtual layer over hardware which allows user to run multiple instances simultaneously on hardware.
- It is a key technology used in cloud computing.
- It is the base on which major cloud computing services such as Amazon EC2, VMware vCloud, etc.
- Hardware virtualization is still one of the most common types of virtualization.

Web 2.0

- It is the interface through which Cloud computing services interact with clients.
- It is because of Web 2.0 that we have interactive and dynamic web pages.
- It also increases flexibility among web pages.
- Popular examples of web 2.0 include Google Maps, Facebook, Twitter, etc.
- social media is possible because of this technology only.
- It gained major popularity in 2004.

Service orientation

- It acts as reference model for cloud computing.
- It supports low-cost, flexible, and evolvable applications.
- 2 important concepts were introduced in this computing model.
- Quality of Service (QoS) , also includes SLA (Service Level Agreement) and Software as a Service (SaaS).

- QoS (Quality of Service) is a networking technique used to manage and prioritize network traffic so that important data gets better performance.

Utility computing

- It is a computing model that defines service provisioning techniques for services such as compute services along with other major services such as storage, infrastructure, etc., which are provisioned on a pay-per-use basis.

NEED FOR CLOUD COMPUTING

- Cloud computing is needed to overcome the limitations of traditional computing systems and to meet modern demands for scalability, flexibility, and cost efficiency.

- **1. Cost Efficiency**
- Traditional IT requires high **capital investment** in hardware, software, and maintenance.
- Cloud computing follows a **pay-as-you-use** model.
- Eliminates expenses for:
  - Physical servers
  - Data centers
  - Cooling and power
  - IT infrastructure maintenance
- **Result:** Lower operational and upfront costs.

- **2. Scalability and Elasticity**
- Organizations face fluctuating workloads.
- Cloud allows **instant scaling up or down** based on demand.
- No need to purchase extra hardware for peak loads.
- **Example:** E-commerce websites during sales events.

- **3. Accessibility and Mobility**
- Cloud services are accessible **anytime, anywhere** via the internet.
- Supports remote work and mobile access.
- Multiple users can collaborate in real time.
- **Example:** Google Docs, Microsoft OneDrive.

NEED FOR CLOUD COMPUTING

- **4. High Availability and Reliability**
- Cloud providers offer:
  - Redundant data storage
  - Backup systems
  - Disaster recovery solutions
- Ensures **minimum downtime** and business continuity.

- **5. Faster Deployment and Time to Market**
- Applications can be deployed in minutes instead of weeks.
- No need for lengthy hardware setup.
- Enables rapid testing, development, and innovation.

- **6. Automatic Updates and Maintenance**
- Cloud providers handle:
  - Software updates
  - Security patches
  - Hardware upgrades
- Reduces workload on IT staff.

- **7. Efficient Resource Utilization**
- Resources are shared among multiple users (multi-tenancy).
- Prevents underutilization of servers.
- Improves energy efficiency.

NEED FOR CLOUD COMPUTING

- **8. Data Backup and Disaster Recovery**
- Automatic data backups across multiple locations.
- Faster recovery in case of:
  - Hardware failure
  - Natural disasters
  - Cyberattacks

- **9. Support for Advanced Technologies**
- Cloud enables easy access to:
  - Artificial Intelligence (AI)
  - Machine Learning (ML)
  - Big Data analytics
  - Internet of Things (IoT)
- No need for specialized local infrastructure.

- **10. Environmental Benefits**
- Optimized data centers consume less energy.
- Reduces carbon footprint compared to individual on-premise systems.

ADVANTAGES OF CLOUD

1. **Cost Efficiency**: Cloud Computing provides flexible pricing to users with principal pay-as-you-go model

2. **Flexibility and Scalability**: Cloud services facilitate scaling of resources based on demand. It ensures efficiency of businesses in handling various workloads

3. **Collaboration and Accessibility**: Cloud computing provides easy access to data and applications from anywhere over internet

4. **Automatic Maintenance and Updates**: Cloud takes care of infrastructure management and keeping with latest software automatically making updates.

5. **It is easier to get backup in cloud:** It allows us **easy** and **quick access** stored information **anywhere** and **Anytime.**

6. It **reduces** both **hardware** and **Software cost,** and it is **easily maintainable.**

DISADVANTAGES OF CLOUD

1. **Security Concerns**

Storing of sensitive data on external servers raised more security concerns which is one of the main drawbacks of cloud computing.

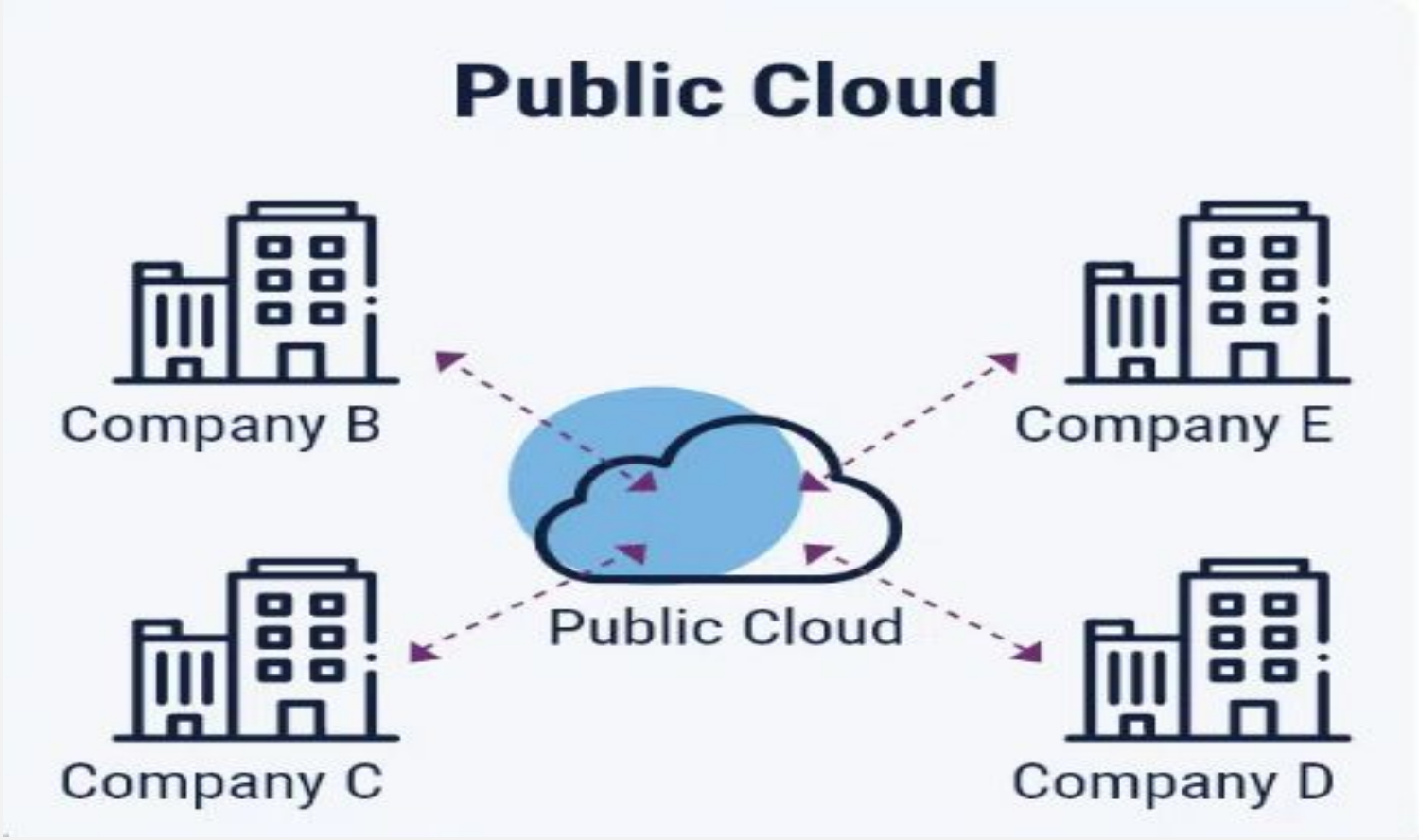
2. **Downtime and Reliability**

Even though cloud services are usually dependable, they may also have unexpected interruptions and downtimes

3. **Dependency on Internet Connectivity**

Cloud computing services heavily rely on Internet connectivity

CLOUD DEPLOYMENT MODELS- PUBLIC CLOUD



CLOUD DEPLOYMENT MODELS- PUBLIC CLOUD

Definition
A **public cloud** is owned and operated by a third-party cloud service provider and shared among multiple users over the internet.

Characteristics
Resources are shared (multi-tenancy)
Accessible via the public internet
Pay-as-you-use pricing
High scalability

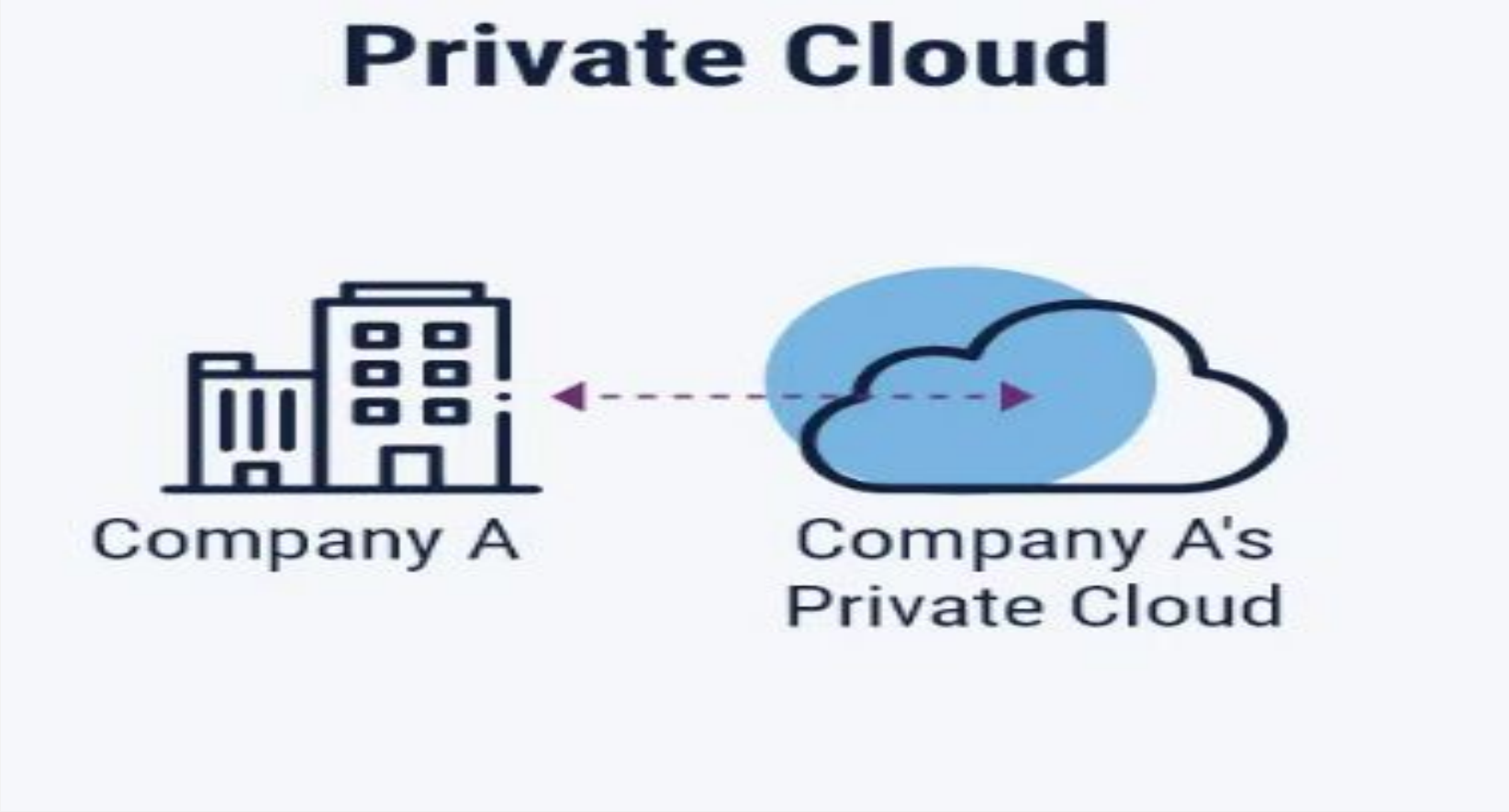
Examples
Amazon Web Services (AWS)
Microsoft Azure
Google Cloud Platform (GCP)

Advantages
Low cost
Easy to use
Highly scalable
No maintenance responsibility for users

Disadvantages
Less control over data
Security and privacy concerns
Dependent on internet connectivity

Use Cases
Web applications
Development and testing
Startups and small businesses

CLOUD DEPLOYMENT MODELS- PRIVATE CLOUD



CLOUD DEPLOYMENT MODELS- PRIVATE CLOUD

Definition
A **private cloud** is dedicated to a single organization and may be hosted on-premises or by a third-party provider.

Characteristics
Single-tenant environment
High security and control
Customizable infrastructure

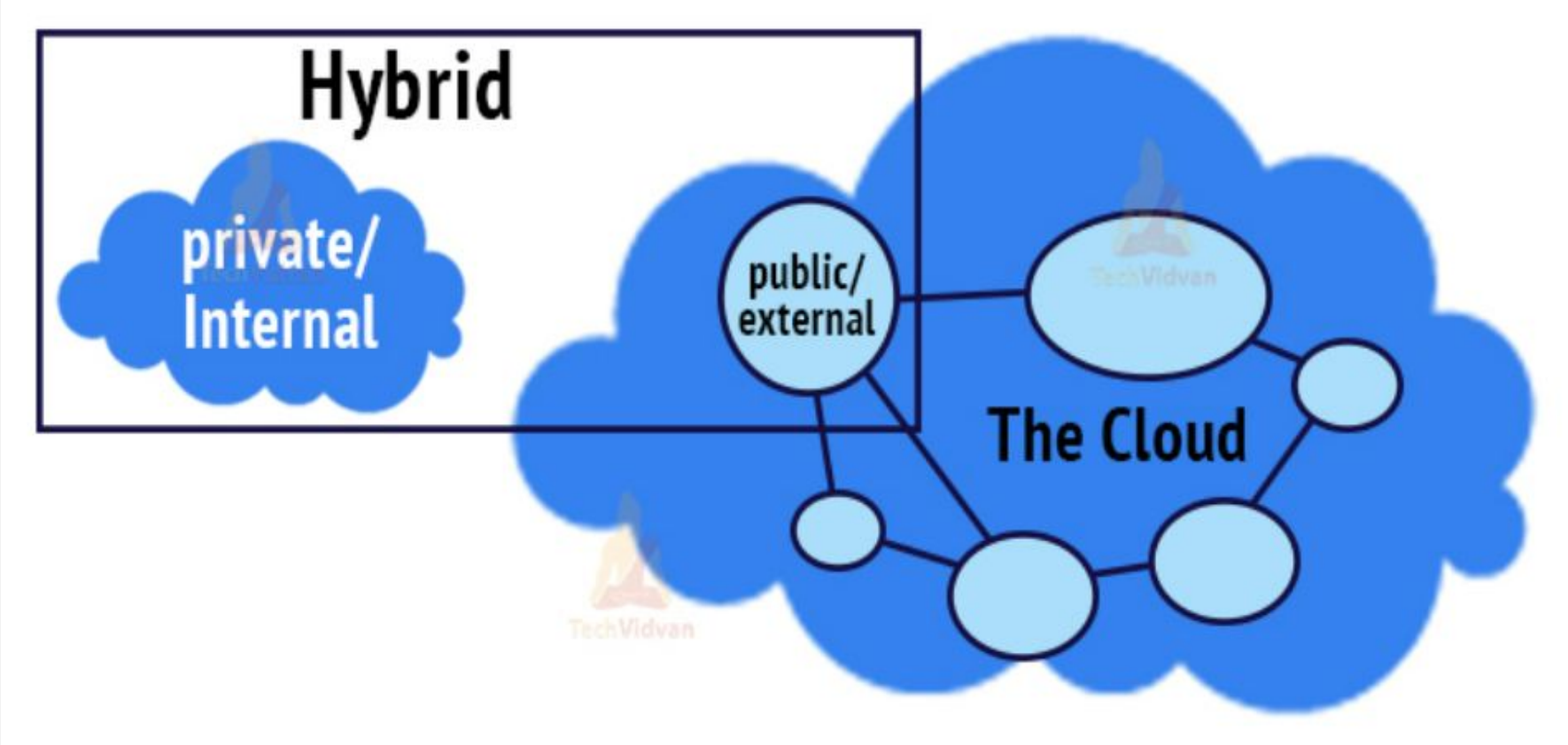
Examples
VMware Private Cloud
OpenStack
Azure Stack

Advantages
Enhanced security and privacy
Full control over data
Better compliance

Disadvantages
High cost
Requires skilled IT staff
Limited scalability compared to public cloud

Use Cases
Banking and financial institutions
Healthcare organizations
Government agencies

CLOUD DEPLOYMENT MODELS- HYBRID CLOUD



CLOUD DEPLOYMENT MODELS- HYBRID CLOUD

Definition
A **hybrid cloud** combines public and private clouds, allowing data and applications to move between them.

Characteristics
Flexible workload placement
Integrated environment
Improved resource optimization

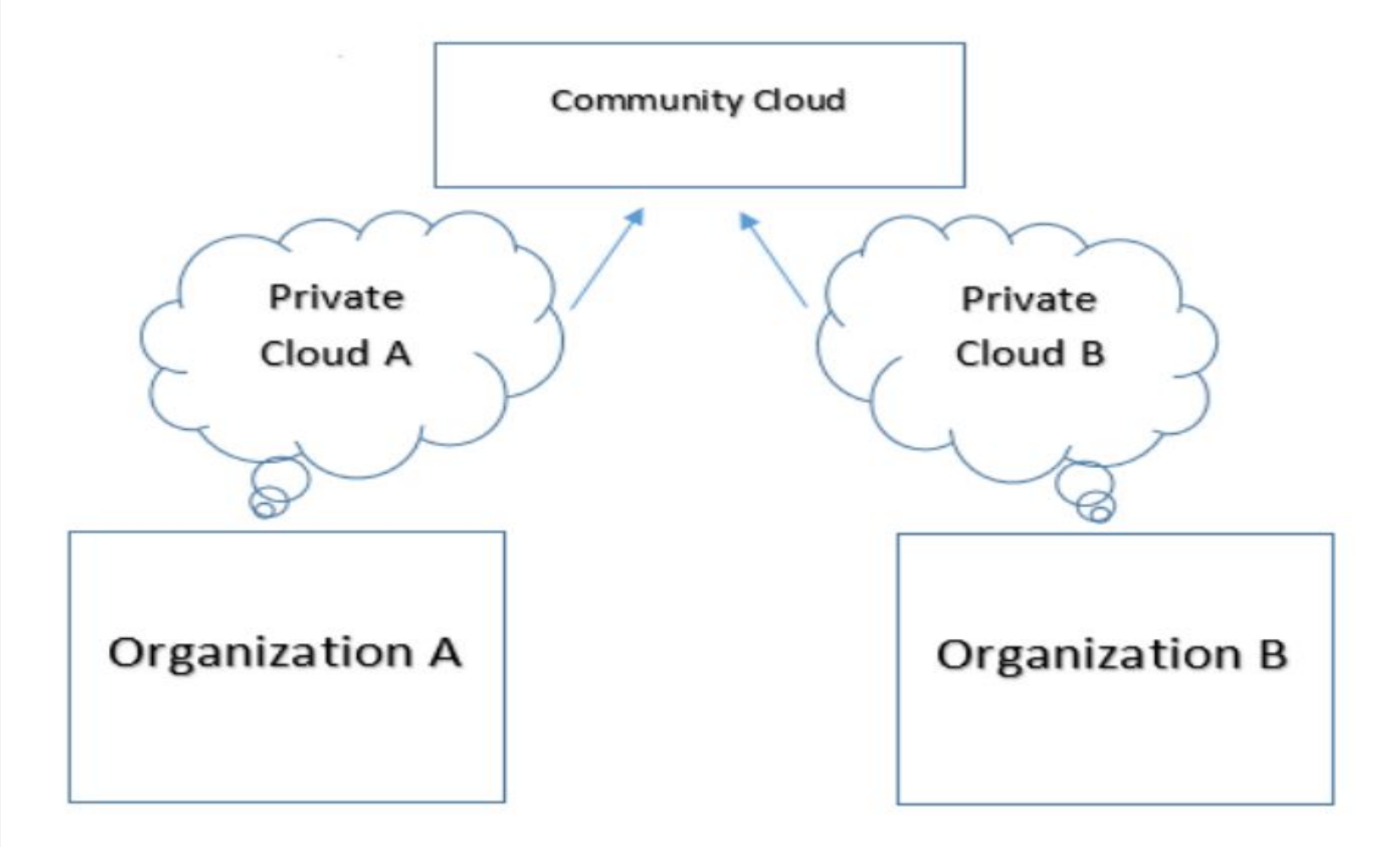
Examples
AWS Outposts
Azure Hybrid Cloud
Google Anthos

Advantages
Cost optimization
Enhanced security
High flexibility

Disadvantages
Complex management
Integration challenges

Use Cases
Data-sensitive applications
Seasonal workloads
Disaster recovery

CLOUD DEPLOYMENT MODELS- COMMUNITY CLOUD



CLOUD DEPLOYMENT MODELS- COMMUNITY CLOUD

Definition
A **community cloud** is shared by multiple organizations with similar requirements, such as security, compliance, or mission objectives.

Characteristics
Shared infrastructure
Common governance policies
Limited user group

Examples
Government community clouds
Education or research clouds

Advantages
Cost sharing
Better compliance than public cloud
Collaboration among organizations

Disadvantages
Limited scalability
Shared resources may cause performance issues

Use Cases
Universities
Research institutions
Government departments

EVOLVING DATA CENTER INTO PRIVATE CLOUD

Traditional Data Centre Limitations

1. Static and siloed infrastructure
2. Underutilized servers and storage
3. Manual provisioning and configuration
4. High capital and operational costs
5. Limited scalability and flexibility
6. Slow response to business needs

Concept of a Private Cloud

A **private cloud** provides cloud-like features such as:

1. On-demand self-service
2. Resource pooling
3. Rapid elasticity
4. Measured service

within a **single organization's environment**.

- Evolving a data centre into a private cloud involves virtualizing compute, storage, and networking resources and adding automation and self-service capabilities. This transformation improves resource utilization, scalability, and efficiency while maintaining security and control.

Architectural Components of a Private Cloud

1. **Compute Layer:** Virtualized servers
2. **Storage Layer:** Virtual storage pools
3. **Network Layer:** Virtual networks and SDN
4. **Management Layer:** Automation, monitoring, security
5. **Security Layer:** Identity management, access control, encryption.

EVOLVING DATA CENTER INTO PRIVATE CLOUD

Server Virtualization
Converts physical servers into **virtual machines (VMs)**.
Multiple VMs run on a single physical server.
Improves resource utilization and isolation.
Technologies: VMware ESXi, Hyper-V, KVM

Storage Virtualization
Abstracts physical storage into a unified pool.
Enables dynamic allocation and scalability.
Supports features like snapshots and replication.

Network Virtualization
Uses software-defined networking (SDN).
Virtual networks, switches, and firewalls are created in software.
Enables secure, flexible network configuration.

Automation and Orchestration
Automates provisioning of compute, storage, and networking.
Reduces manual intervention.
Improves speed and consistency.
Tools: OpenStack, VMware vRealize, Azure Stack

Self-Service Portal
Users can request resources on demand.
IT acts as a service provider.
Improves user experience and efficiency.

Resource Management and Monitoring
Tracks usage, performance, and availability.
Enables capacity planning and optimization.
Supports chargeback or showback models.

DATA CENTRE COMPONENTS

A **data center** is a facility that houses computing systems and related components used for **data processing, storage, and communication**.

Compute Components (Servers)
Physical machines that run applications and process data.
Types:
- Rack servers
- Blade servers

Often virtualized using hypervisors.
Purpose: Processing and computation.

Storage Components
Used for persistent data storage.
Types:
- **DAS (Direct Attached Storage)**
- **NAS (Network Attached Storage)**
- **SAN (Storage Area Network)**

Storage media includes HDDs, SSDs, and hybrid systems.
Purpose: Data storage and backup.

DATA CENTRE COMPONENTS

Networking Components
Enable communication within the data center and with external networks.
Components include:
- Switches
- Routers
- Load balancers
- Firewalls

Purpose: Data transfer and connectivity.

Power Infrastructure
Supplies uninterrupted electrical power.
Includes:
- Uninterruptible Power Supply (UPS)
- Power Distribution Units (PDUs)
- Backup generators

Purpose: Ensure continuous operation.

Cooling and Environmental Control
Maintains temperature and humidity.
Components include:
- Air conditioning units (CRAC/CRAH)
- Cooling fans
- Chillers

Purpose: Prevent overheating and equipment failure.

Physical Infrastructure
Structural elements of the data center.
Includes:
- Racks and cabinets
- Raised floors
- Cabling systems

Purpose: Equipment housing and airflow management.

DATA CENTRE COMPONENTS

Security Components
Physical Security
CCTV cameras
Biometric access systems
Security personnel
Fire detection and suppression systems
Logical Security
Firewalls
Intrusion Detection/Prevention Systems (IDS/IPS)
Access control mechanisms
Purpose: Protect data and infrastructure.

Management and Monitoring Systems
Data Center Infrastructure Management (DCIM)
Performance monitoring tools
Alerting and logging systems
Purpose: Resource management and fault detection.

Backup and Disaster Recovery Components
Backup servers and storage
Replication systems
Off-site disaster recovery locations
Purpose: Data protection and business continuity.

Software Components
Operating systems
Virtualization platforms
Cloud management and automation tools
Purpose: Control and optimize data center operations.

CLOUD SECURITY

- **Cloud security** refers to the set of policies, controls, technologies, and best practices designed to protect cloud-based systems, data, and infrastructure.

Need for Cloud Security

1. Data is stored on remote servers.
2. Shared (multi-tenant) environments increase risk.
3. Threats include data breaches, data loss, and cyberattacks.
4. Compliance and privacy requirements must be met.

Cloud Security Objectives

1. **Confidentiality:** Prevent unauthorized data access
2. **Integrity:** Protect data from alteration
3. **Availability:** Ensure services remain accessible
4. (Also known as the **CIA Triad**)

Shared Responsibility Model

- **Cloud Provider Responsibility:**
  1. Physical data centre security
  2. Infrastructure security
  3. Hardware and networking
- **Customer Responsibility:**
  1. Data security
  2. User access management
  3. Application security
  4. Configuration settings

CLOUD SECURITY- CLOUD SECURITY CONTROLS

Identity and Access Management (IAM)
User authentication and authorization
Role-based access control (RBAC)
Multi-factor authentication (MFA)

Data Security
Encryption at rest and **in transit**
Data masking and tokenization
Secure key management

Network Security
Firewalls and security groups
Virtual private networks (VPNs)
Intrusion Detection and Prevention Systems (IDS/IPS)

Application Security
Secure application design
Regular patching and updates
Secure APIs

Monitoring and Logging
Continuous security monitoring
Audit logs and alerts
Threat detection and incident response

CLOUD SCALABILITY

Cloud scalability refers to the ability of a cloud system to **increase or decrease computing resources** (such as CPU, memory, storage, and bandwidth) according to workload demand.

Need for Cloud Scalability:

1. Workloads are not constant.
2. Traditional systems require fixed capacity.
3. Peak loads cause performance issues.
4. Over-provisioning leads to resource wastage.
5. **Cloud scalability solves these problems by providing resources on demand.**

CLOUD SCALABILITY- TYPES

Vertical Scalability (Scale Up / Scale Down)
Increasing or decreasing the capacity of a single resource.
Example: Adding more RAM or CPU to a server.
Limited by hardware capacity.

Advantages
Simple to implement
No application redesign required

Disadvantages
Hardware limitations
Downtime may be required

Horizontal Scalability (Scale Out / Scale In)
Adding or removing multiple servers or instances.
Load is distributed across systems.
Most commonly used in cloud environments.

Advantages
High availability
No single point of failure
Better performance

Disadvantages
Requires load balancing
More complex architecture

TIME TO MARKET

Time to market (TTM) refers to the total time taken to develop, test, deploy, and make a product or service available to users. Cloud computing significantly reduces time to market compared to traditional IT systems.

Need for Reduced Time to Market

1. Competitive business environment
2. Rapidly changing customer requirements
3. Demand for faster innovation
4. Traditional infrastructure setup is slow

HOW CLOUD COMPUTING REDUCES TIME TO MARKET

On-Demand Infrastructure
Servers, storage, and networks are available instantly.
No need to procure or install hardware.
Resources can be provisioned in minutes.

Faster Application Development
Ready-to-use platforms and services.
Developers focus on coding instead of infrastructure.
Pre-configured environments speed up development.

Automation and DevOps Support
Continuous Integration and Continuous Deployment (CI/CD).
Automated testing and deployment pipelines.
Faster release cycles.

Scalability and Elasticity
Applications scale automatically after deployment.
No redesign needed to handle increased users.
Ensures smooth launch and growth.

Rapid Testing and Experimentation
Easy creation of development and test environments.
Supports quick prototyping and proof of concepts (PoC).
Failed experiments can be discarded at low cost.

DISTRIBUTION OVER THE INTERNET

- In cloud computing, **distribution over the internet** refers to the practice of **spreading computing resources, data, and services across multiple networked systems** that communicate via the internet to deliver scalable, reliable, and efficient services.

Meaning of Distribution
Computing tasks are not handled by a single machine.
Workloads are divided and executed across **multiple servers and locations**.
Users access these resources through the internet without knowing their physical location.

Role of the Internet in Cloud Computing
Acts as the primary communication medium.
Connects users to cloud data centers globally.
Enables remote access to applications, storage, and services.

DISTRIBUTION OVER THE INTERNET

Key Components of Distributed Cloud Systems

Distributed Computing
Tasks are split into smaller units.
Executed in parallel across different servers.
Improves performance and speed.

Distributed Storage
Data is stored across multiple physical locations.
Ensures redundancy and fault tolerance.
Enhances availability and durability.

Distributed Networking
Load balancers route traffic to optimal servers.
Reduces latency and avoids congestion.
Supports high availability.

Distribution Models in Cloud Computing

Geographical Distribution
Data centers are spread across regions.
Improves latency and disaster recovery.

Logical Distribution
Virtual machines and containers distributed logically.
Independent of physical hardware.

CLOUD COMPUTING CASE STUDIES- ASSIGNMENT QUESTIONS

Case Study: Online Learning Platform
An online learning platform faces rapid user growth during exam seasons.
Identify the cloud services and deployment model best suited.
Explain how cloud scalability and load balancing ensure performance.

Case Study: Banking Sector
A bank plans to migrate its core applications to the cloud but has strict security requirements.
Recommend a suitable cloud deployment model.
Discuss security measures needed to protect sensitive data.

Case Study: Startup Application Development
A startup wants to develop, test, and deploy applications quickly with minimal cost.
Suggest suitable cloud service models.
Justify how cloud computing supports rapid innovation.

Case Study: Disaster Recovery
An organization wants a cost-effective disaster recovery solution.
Explain how cloud-based disaster recovery works.
Discuss its advantages over traditional backup systems.